

Arrhythmias — Polymorphic VT, Torsades & VF



1. Torsades twisting axis

WHAT YOU SEE

Tornado 'TORSADES' with 'AXIS IS WAVES... continuously shifting & rotating'

WHAT IT ENCODES

Torsades de pointes ('twisting of the points') is a polymorphic VT in which the QRS amplitude and axis sinusoidally twist around the isoelectric baseline, typically at rates of 200–250 bpm. The hallmark is that it arises specifically in the setting of a prolonged QT interval and is classically pause-dependent (initiated by a short-long-short R-R sequence). Recognize the picture: long QT, then a PVC on the T wave (R-on-T) launching the twisting tachycardia.

BOARD TRAP

WRONG: torsades has a fixed monomorphic QRS — that describes scar-related monomorphic VT. The defining feature of torsades is the continuously CHANGING (twisting) QRS axis on a background of a long QT; a uniform QRS rules it out.

2. Prolonged QT

WHAT YOU SEE

Stretched ruler 'PROLONGED QT'

WHAT IT ENCODES

Torsades is driven by QT prolongation (QTc >500 ms markedly raises risk); recognizing a long QT before the arrhythmia appears is the key board cue. QT prolongs repolarization, allowing early afterdepolarizations to trigger the rhythm. Always use the corrected QT (QTc, Bazett) since the raw QT shortens with faster rates.

BOARD TRAP

WRONG: judging the raw QT instead of the rate-corrected QTc. A normal-looking raw QT at a slow rate can still be a dangerously prolonged QTc — always correct for heart rate before excluding long-QT–driven torsades.

3. IV magnesium

WHAT YOU SEE

Star 'IV MAGNESIUM'

WHAT IT ENCODES

IV magnesium sulfate (2 g IV bolus over 1–2 min, repeat in 5–15 min, then infusion) is first-line treatment for torsades EVEN when the serum magnesium is normal — it stabilizes the myocardial membrane and suppresses early afterdepolarizations. Mnemonic: 'Magnesium for the twist.' Combine with stopping offending drugs and correcting K+.

BOARD TRAP

WRONG: giving amiodarone or other class III/la antiarrhythmics — these FURTHER prolong the QT and worsen torsades. The reflex 'amiodarone for VT' answer is a classic trap; for torsades the answer is magnesium, not amiodarone.

4. Stop QT-prolonging drugs

WHAT YOU SEE

List 'CLASS Ia / ANTIPSYCHOTICS / MACROLIDES / FLUOROQUINOLONES'

WHAT IT ENCODES

Acquired long QT: immediately stop the offending agents — class Ia (quinidine, procainamide, disopyramide) and class III (sotalol, dofetilide) antiarrhythmics, antipsychotics (haloperidol, ziprasidone), antiemetics (ondansetron), macrolides (azithromycin, erythromycin), fluoroquinolones (levofloxacin), and methadone. Mnemonic 'ABCDE' — Antiarrhythmics, antiBiotics, antipsyChotics/antiemetics, antiDepressants, Electrolytes. Drug-induced QT prolongation is the most common cause of torsades.

BOARD TRAP

WRONG: continuing a 'medically necessary' QT-prolonging drug because the patient needs it — once torsades occurs the offending drug must be stopped. Watch for additive risk when two QT-prolonging drugs (e.g., a macrolide plus an antipsychotic) are co-prescribed.

5. Correct K / Mg / Ca

WHAT YOU SEE

Gauges 'K+ Mg2+ Ca2+'

WHAT IT ENCODES

Aggressively correct hypokalemia (target K+ 4.5–5.0 mEq/L), hypomagnesemia, and hypocalcemia — each prolongs repolarization and precipitates torsades. Diuretic use, vomiting/diarrhea, and refeeding are common culprits. Replete potassium and magnesium together since hypomagnesemia causes refractory hypokalemia.

BOARD TRAP

WRONG: repleting potassium alone while ignoring magnesium — low Mg2+ drives renal potassium wasting, so K+ will not correct until Mg2+ is replaced. Always give magnesium alongside potassium.

6. EAD trigger

WHAT YOU SEE

Label 'EAD'

WHAT IT ENCODES

Early afterdepolarizations (EADs) are abnormal depolarizations during phase 2/3 of the action potential, occurring when prolonged repolarization reactivates L-type calcium channels. EADs are the trigger that initiates torsades on a long-QT substrate. They are favored by bradycardia and pauses, which is why pause-dependent torsades responds to overdrive pacing.

BOARD TRAP

WRONG: confusing EADs (early, cause torsades/long-QT) with delayed afterdepolarizations (DADs, occur after full repolarization, caused by calcium overload as in digoxin toxicity and CPVT). EAD = long QT torsades; DAD = digoxin/catecholamine triggered activity.

7. Congenital LQTS

WHAT YOU SEE

DNA helix 'CONGENITAL LQTS'

WHAT IT ENCODES

Congenital long-QT syndromes are inherited channelopathies causing QT prolongation and torsades, distinct from acquired/drug causes. LQT1 (KCNQ1, IKs) triggers with exercise/swimming and is treated with beta-blockers; LQT2 (KCNH2/hERG, IKr) triggers with emotional stress/auditory stimuli; LQT3 (SCN5A, late INa) triggers during sleep/rest. Beta-blockers are mainstay; ICD for high-risk or post-arrest.

BOARD TRAP

WRONG: giving a beta-blocker is universally protective — in LQT3 (sodium-channel, rest-triggered) beta-blockers are less effective and excessive bradycardia can be harmful. Match trigger to subtype: exercise→LQT1, startle/emotion→LQT2, sleep→LQT3.

8. Overdrive pacing

WHAT YOU SEE

'OVERDRIVE PACING'

WHAT IT ENCODES

For recurrent pause-dependent torsades, increasing the heart rate shortens the QT and abolishes the pauses that trigger EADs. Use temporary transvenous overdrive pacing (target ~90–110 bpm) or IV isoproterenol as a pharmacologic bridge to raise the rate when magnesium alone is insufficient. Treat any underlying bradycardia/AV block.

BOARD TRAP

WRONG: using isoproterenol in CONGENITAL long-QT — raising catecholamines can worsen congenital LQTS (especially LQT1/2). Isoproterenol/overdrive pacing is for ACQUIRED pause-dependent torsades; congenital LQTS is managed with beta-blockers.

9. Self-terminates

WHAT YOU SEE

'SELF-TERMINATES'

WHAT IT ENCODES

Torsades episodes are often brief and self-terminate, producing syncope or near-syncope; however, runs can be sustained and degenerate into ventricular fibrillation and sudden cardiac death. Recurrent self-terminating episodes still mandate urgent treatment (magnesium, correct triggers) because the next run may not stop.

BOARD TRAP

WRONG: a self-terminating episode is benign and needs no treatment — recurrent torsades carries a high risk of degenerating to VF/SCD and must be treated emergently even between episodes.

10. VF → sudden death

WHAT YOU SEE

'VF → SUDDEN DEATH'

WHAT IT ENCODES

Polymorphic VT/torsades can deteriorate into ventricular fibrillation, the immediate mechanism of sudden cardiac death. Once VF or pulseless polymorphic VT occurs, the patient is in cardiac arrest requiring CPR and defibrillation. This is why prompt correction of the QT substrate is preventive, not merely supportive.

BOARD TRAP

WRONG: treating a now-pulseless torsades patient with more magnesium or antiarrhythmics first — once it degenerates to VF/pulseless VT the answer is immediate unsynchronized defibrillation plus CPR, not drugs.

11. Immediate defibrillation

WHAT YOU SEE

AED 'IMMEDIATE DEFIBRILLATION — NO DRUGS FIRST'

WHAT IT ENCODES

Pulseless VT and VF (and hemodynamically unstable polymorphic VT that has no reliable pulse) require immediate UNSYNCHRONIZED defibrillation (200 J biphasic), not synchronized cardioversion and not drugs first. Synchronization fails in polymorphic/chaotic rhythms because there is no consistent R wave to sense. Resume CPR immediately after the shock.

BOARD TRAP

WRONG: attempting SYNCHRONIZED cardioversion for pulseless or polymorphic VT — the defibrillator cannot reliably sync to the irregular QRS and may withhold the shock. Polymorphic/pulseless → defibrillate (unsynchronized); organized monomorphic VT with a pulse → synchronized cardioversion.

12. Most common cause = ischemia

WHAT YOU SEE

Banner 'MOST COMMON CAUSE = ISCHEMIA'

WHAT IT ENCODES

Polymorphic VT with a NORMAL baseline QT is most often caused by acute myocardial ischemia/infarction — the treatment is revascularization, anti-ischemic therapy, and beta-blockade, NOT magnesium. Distinguish from torsades, which by definition has a long QT. Checking the inter-episode QT is the discriminator that decides therapy.

BOARD TRAP

WRONG: labeling all polymorphic VT as torsades and reaching for magnesium — polymorphic VT with a normal QT is ischemic and needs revascularization/beta-blockade. The QT interval between episodes is the key board discriminator (long QT → torsades/magnesium; normal QT → ischemia).

13. Hyperkalemia / Na-blocker toxicity

WHAT YOU SEE

Banana 'HYPERKALEMIA' and 'Na BLOCKER TOXICITY'

WHAT IT ENCODES

A wide, slow, sinusoidal-appearing wide-complex rhythm should prompt thought of hyperkalemia (peaked T waves → widened QRS → sine wave → arrest; treat with IV calcium gluconate, insulin/glucose, then shift/eliminate K+) and sodium-channel-blocker toxicity (TCAs, class Ia/Ic, cocaine; wide QRS and terminal R in aVR — treat with IV sodium bicarbonate boluses). Both are reversible mimics of VT.

BOARD TRAP

WRONG: defibrillating or loading antiarrhythmics for a sine-wave rhythm without checking potassium/drug history — hyperkalemia needs IV calcium and shifting therapy, and Na-channel-blocker toxicity needs sodium bicarbonate. Class I antiarrhythmics would worsen sodium-blocker toxicity.

14. Ventricular flutter

WHAT YOU SEE

Red continuous sine-wave rhythm strip 'VENTRICULAR FLUTTER' along the bottom

WHAT IT ENCODES

Ventricular flutter is a very rapid (~300 bpm) regular ventricular rhythm appearing as a continuous sinusoidal/zigzag wave with no clear distinction between QRS and T wave. It is hemodynamically unstable, behaves like VF, and rapidly degenerates into VF and cardiac arrest. Management is immediate defibrillation.

BOARD TRAP

WRONG: attempting synchronized cardioversion or drug therapy for ventricular flutter — at ~300 bpm with a continuous sine-wave morphology there is no reliable R wave to synchronize to, so treat as VF with immediate unsynchronized defibrillation.

15. Monomorphic vs polymorphic board

WHAT YOU SEE

'WEATHER COMPARISON BOARD' — MONOMORPHIC (same each beat / normal QT) vs POLYMORPHIC (prolonged QT / no QRS)

WHAT IT ENCODES

The comparison board is the core board discriminator: MONOMORPHIC VT = identical QRS each beat, arises from scar/structural disease, normal QT — treat with cardioversion (unstable) or amiodarone/procainamide (stable). POLYMORPHIC VT = beat-to-beat varying QRS; if QT is long it is torsades (→ magnesium), if QT is normal it is usually ischemic (→ revascularize/beta-block).

BOARD TRAP

WRONG: choosing therapy off morphology alone without the QT — polymorphic VT splits by QT (long→magnesium/torsades; normal→ischemia). Reading the baseline QT is what separates the two polymorphic pathways and determines the correct answer.
